

1 Remaining Run Time

Definition

The remaining run time is the amount of time that remains to process an operation, accounting for any materials or parts that have already been processed or produced.

If the quantity that has already been produced is 0, the remaining run time is equal to the processing time for the operation.

Using the remaining run time option in HYDRA shop floor scheduling

The remaining run time is calculated for HYDRA shop floor scheduling based on a formula stored for an operation. To accomplish this, a reference to a formula must be defined in formula management in the field "Formula RRT1" at the operation. The formula is either defined manually via the corresponding update function or it is transferred at the PPS interface. If no customer specific formulas have been defined, enter the value "RRT" here. In this case, the remaining run time is calculated based on the "Target cycle / 1,000 / TLG * (target quantity - yield produced up until now)". Care should be taken to ensure that the fields target cycle and partitioning have been filled in correctly for the operation.

As a rule: The display is not updated (showing the new calculation of the remaining run time) in the graphic planning board until the data has been requested again.

The field "Remaining run time 2" is provided for a second, alternative remaining run time; it is also displayed in the order overview. This formula is not used in the HYDRA shop floor scheduling module.

Error! Reference source not found.Cycle/ quantity-based remaining run times

The default remaining run time formula RRT set from the factory calculates the remaining run time based on the target cycle defined for the operation:

```
(( (ANR.SZY/1000) * (ANR.SGR:GUTP-ANR.EGR:GUTP) / ANR.TLG) * (ANR.SGR:GUTP-ANR.EGR:GUTP>0))
```

If the remaining run time should be calculated based on the operation's actual cycle, then a new remaining run time formula must be defined for this:

```
(( ((ANR.IZY > 0) * ANR.IZY + (ANR.IZY <= 0) * ANR.SZY) / 1000) * (ANR.SGR:GUTP-ANR.EGR:GUTP) / ANR.TLG) * (ANR.SGR:GUTP-ANR.EGR:GUTP>0)
```

The formula accounts for the fact that the target cycle is used for operations with an actual cycle 0 (e.g. for operations with status "prepared").

Time-based remaining run time

If the remaining run time should not be calculated based on the produced quantities, but instead based on the (posted) time that has already elapsed, this can be done, for example, using the following formula:

$(\text{ANR.BEARBZ} - \text{ANR.EGR:BMK11}) * (\text{ANR.BEARBZ} - \text{ANR.EGR:BMK11} > 0)$

The value ANR.BEARBZ uses the value "processing time" from the order backlog (auftrags_bestand table), whereas the value ANR.EGR:BMK11 is drawn from the order status (auftrag_status table). Time/duration progresses in the order status (auftrag_status table) in the event of a

- Manual status change (M_MST)
- OP interrupt (A_UN)
- OP logoff (A_AB)
- Partial confirmation (A_TR)
- Log-ons or log-offs by a person (P_AN, P_AB)
- Any auto status (M_AST), i.e. approx. every 120 seconds

The last option (M_AST) requires that the workplace is configured as an MDE workplace, i.e. it is assigned to a terminal that is running in operation mode MDE (defined in the terminal configuration or alternately in the terminal assignment). Therefore, where ADE workplaces are concerned, the times/durations do not progress.